EVRYN Multi-Stage Causal Extraction Workflow & Modular Verification Architecture

STAGE 1 — DELIVERED

Candidate Generation

- Bio-Entity Span Detection
- Intra-Sentence Entity Pairs
- Paragraph & Section Bounds
- Pre-flight Quality Gating

STAGE 2 — DELIVERED

Syntactic Validation

- Forward Causal Triggers
- Passive Inversion Rules
- Agent-Nominal Patterns
- Strict Negation Quarantining

STAGE 3 — DELIVERED

Context Resolution

- Demonstrative Anaphora
- Abbreviation Normalization
- Discourse Connectives
- Dual-Sentence Alignment

STAGE 4 — 0.5350 F1

Multi-Page Linking

- Bounded Multi-Hop (<= 2 hops)
- Domain-Invariant Pruning
- Reverse Regulator Gating
- Confidence Compounding

Direction Check

Dual Spans [Sc, Se]

Transitive Paths

Strict False-Positive Controls & Domain Invariants — ACTIVE INVARIANTS

- Terminal Phenotype Invariant: Virulence and survival cannot cause upstream regulatory genes
- Environmental Invariant: Extracellular conditions (pH, temp) cannot be transcriptionally activated
- Homology & Method Gate: Non-causal cloning, homologue analogies, and methods suppressed
- Relative Clause Barrier: Suppresses spurious cross-clause entity-trigger associations
- Master Regulator Directionality: Restricts reverse feedback loops (InvF -> HilA, EspA -> PhoP)

Gated

5. Provenance Grounding & Verification — VERIFIED (100%)

- Dual-Sentence Evidence Tracking: Verbatim [S_causal, S_entity] bindings with zero hallucination
- Calibrated Confidence Metric: Empirical Platt-scaled posterior precision score in [0.0, 1.0]
- Full Text Auditability: Exact character indices [c_start, c_end] and physical page numbers
- Selective Prediction Abstention: Rejection threshold tau = 0.45 for uncertain assertions
- Cryptographic Integrity: Frozen SHA-256 ground-truth ledger across all 20 benchmark papers